Drug Safety (2025) 48:1229–1241
https://doi.org/10.1007/s40264-025-01569-y

ORIGINAL RESEARCH ARTICLE

Large-scale Empirical Identification of Candidate Comparators for Pharmacoepidemiological Studies

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Accepted: 23 May 2025 / Published online: 4 June 2025
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Abstract

Background and Objective The new user cohort design has emerged as a best practice for the estimation of drug effects from observational data. However, despite its advantages, this design requires the selection and evaluation of comparators for appropriateness, a process that can be challenging. The objective of this work was to introduce an empirical approach to rank candidate comparators in terms of their similarity to a target drug in high-dimensional covariate space.

Methods We generated new user cohorts for each RxNorm ingredient and Anatomic Therapeutic Chemical level 4 class in five administrative claims databases then extracted aggregated pre-treatment covariate data for each cohort across five clinically oriented domains. We formed all pairs of cohorts with ≥ 1000 patients and computed a scalar similarity score, defined as the average of cosine similarities computed within each domain, for each pair. We then generated ranked lists of candidate comparators for each cohort.

Results Across up to 1350 cohorts forming 922,761 comparisons, drugs that were more similar in the Anatomic Therapeutic Chemical hierarchy had higher cohort similarity scores. The most similar candidate comparators for each of six example drugs corresponded to alternative treatments used in the target drug's indication(s), and choosing the top-ranked comparator for randomly selected drugs tended to produce balance on most covariates. This approach also ranked highly those comparators chosen in high-quality published new user cohort design studies.

Conclusion Empirical comparator recommendations may serve as a useful aid to investigators and could ultimately enable the automated generation of new user cohort design-derived evidence, a process that has previously been limited to self-controlled designs.

1 Introduction

Pharmacoepidemiologic studies routinely employ cohort-based designs to estimate drug effects using observational data. Although many variations on the cohort study design exist, the new user (NU, sometimes called "incident" new user or "initiator") design [1, 2] has emerged over the past two decades as a best practice. This design, which is intended to emulate a randomized trial's well-defined, exposure-based timescale, involves identifying cohorts of patients initiating one of two drugs: a target drug and a comparator. Patients are then followed for the occurrence of a study outcome, and the incidence is compared

Key Points

- Empirical comparator recommendations based on similarity to a target cohort's covariate profile can be generated rapidly from real-world data.
- These recommendations align well with subject matter knowledge, published literature, and existing drug hierarchies and are correlated with more traditional metrics of covariate balance such as the standardized mean difference.
- Investigators can review empirical comparators from five databases through a public web application or by implementing our methodology against their own data using publicly available code.

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between groups, usually with adjustment for confounding factors. The NU design can be further divided based on whether an active comparator (i.e., an alternative treatment for the indication of interest) or an inactive comparator (i.e., something not used for the indication of interest and which is believed to be unrelated to the outcome of interest) is chosen [3]. The primary advantage of the NU approach is the mitigation of biases known to impact comparisons against nonuser (i.e., unexposed or untreated) groups. In particular, healthy user bias, healthy adherer bias, and immortal time bias may all be reduced by the adoption of an NU design, with particular reductions in confounding by indication expected if an active comparator is chosen [4, 5].

However, although the NU design has many advantages, it requires the study designer to select an appropriate comparator, and the process of identifying and determining comparator adequacy can be challenging. The ideal comparator group represents a counterfactual outcome experience determined by the research question of interest. This question may concern whether the incidence of an outcome in the presence of the target exposure is different than what would have occurred if some other exposure were substituted for the target, or—alternatively—whether that incidence differs from what would have occurred if the target exposure were simply removed. In the former case, which is termed comparative effectiveness or safety research, investigators have historically selected comparators on the basis of perceived clinical equipoise and empirical feasibility. In the latter case, which is common in the drug safety surveillance context, the use of a comparator otherwise thought to be wholly unrelated to the study outcome has been recommended [6–9] and a theoretical framework for its choice developed [10]. Although the counterfactual represented by the comparator thus varies according to the research context, a common goal is to identify a comparator with a similar patient composition in terms of demographics and relevant medical characteristics to the target. Furthermore, although several sets of recommendations [11] have emerged to guide investigators in the comparator selection process, it remains at least partially subjective and difficult to scale in a context with many target drugs. With thousands of drugs available on the market, each of which could, at least in theory, be a viable comparator for another, the selection of comparators for all drugs is thus a large task encompassing nearly 1 million target–comparator possibilities.

We therefore sought to develop an empirical approach for ranking candidate comparators that reflects clinical and epidemiological thinking regarding comparator selection, with the goal of aiding investigators in the study design process. In this study, we present a method for ranking comparators based on their similarity to target drugs (or drug classes) in a high-dimensional covariate space. Additionally, we evaluated the performance of this method by means of comparison against existing drug classification systems, subject matter knowledge, and published literature and describe how it relates to more familiar epidemiological measures of cohort similarity.

2 Methods

Our method for ranking candidate comparators has three stages: the definition and generation of NU cohorts, the extraction of aggregate, pre-treatment covariate data for each cohort, and the computation of similarity (Fig. 1). These stages are described in further detail in the following sections.

2.1 Data Sources

We utilized data from five sources of insurance claims: Merative™ MarketScan® Commercial Claims and Encounters (CCAE), Merative™ MarketScan® Multi-State Medicaid, Merative™ MarketScan®Medicare Supplemental, Japan Medical Data Center (JMDC), and Optum© De-Identified Clinformatics© Data Mart. All data were mapped to the Observational Medical Outcomes Partnership Common Data Model [12]. Episodes of continuous exposure to individual RxNorm drug ingredients were inferred from pharmacy claims [13]. International Classification of Diseases diagnosis codes were mapped to the Systematized Nomenclature of Medicine (SNOMED) Clinical Terms.

2.2 Cohort Definitions

In each data source, we defined cohorts of NUs for each RxNorm ingredient and each Anatomic Therapeutic Chemical (ATC) level 4 class. Over 10,000 ingredient concepts are included in the RxNorm vocabulary and nearly 1000 ATC level 4 classes exist, but far fewer of each are routinely used in medical practice, and the number varies according to local usage patterns in each data source. To enter a given NU cohort, patients were required to have 365 days of continuous observation in the database prior to the first observed exposure to that ingredient or class, which served as their index date. Patients were allowed to enter a given cohort only once, but there was no restriction on the number of distinct cohorts they could enter. To ensure robust estimation of covariate prevalence, we restricted analysis to cohorts with at least 1000 patients.

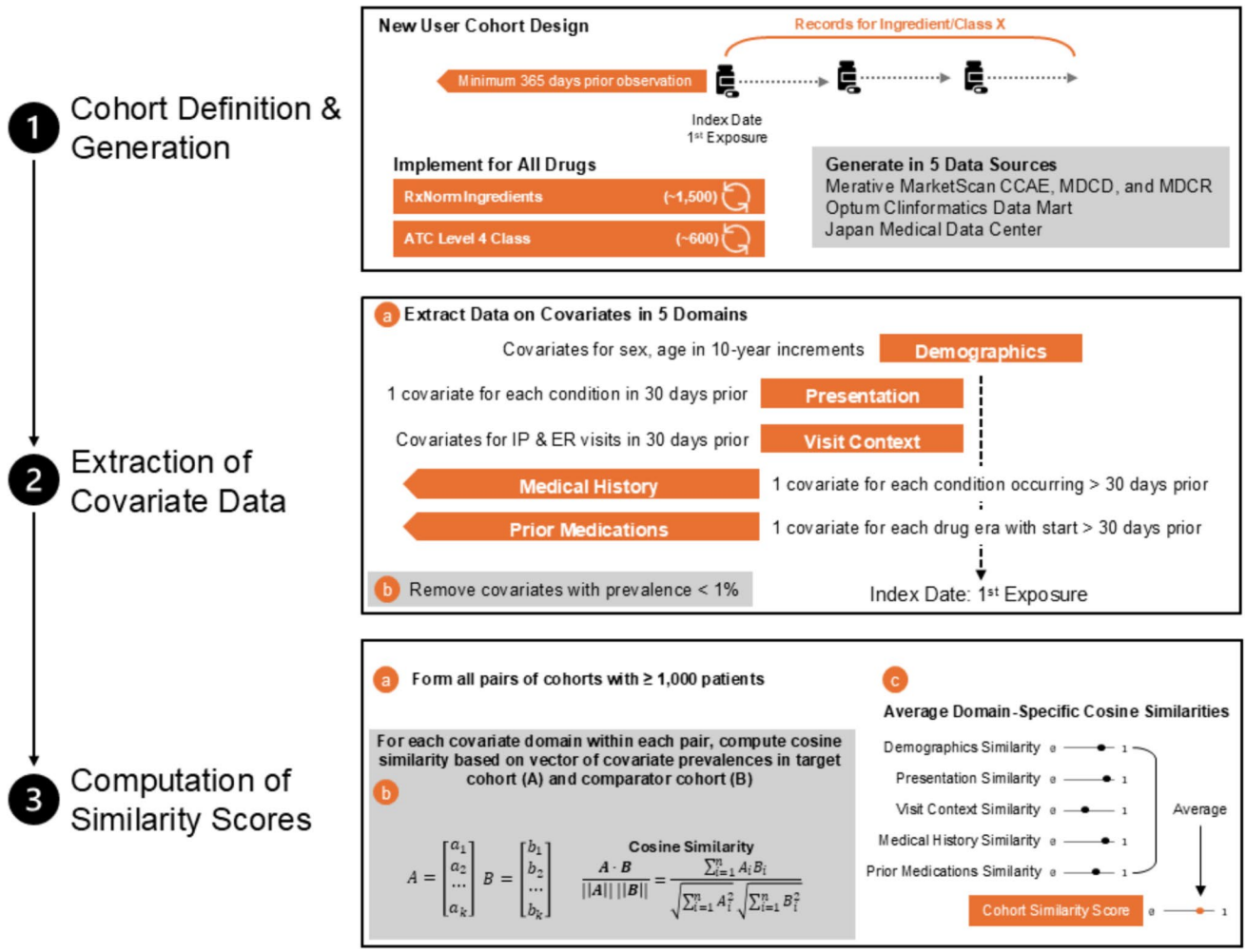

Fig. 1 Depiction of analytic workflow. Step 1: cohort definition and bulk generation in real-world databases, with approximate numbers of ingredient- and class-based cohorts. Step 2: (a) extraction of covariate data for each cohort, (b) removal of rare covariates. Step 3: (a) formation of all pairs of cohorts with at least 1000 patients, (b) computation of cosine similarity using covariate prevalence data in each domain, (c) averaging of domain-specific cosine similarity scores to obtain final cohort similarity score. *ATC* Anatomic Therapeutic Chemical, *CCAE* Commercial Claims and Encounters, *ER* emergency room, *IP* inpatient, *MDCD* Multi-State Medicaid, *MDCR* Merative™ MarketScan®Medicare Supplemental

2.3 Extraction of Covariate Data

Within each cohort, a number of binary covariates were defined across five domains intended to capture the clinical thinking surrounding a treatment decision. These domains were as follows:

- **Demographics**: one covariate defined for female sex, one for male sex, and one covariate for each decade of age
- **Medical history**: one covariate defined for each SNOMED condition observed in the period from the start of patient observation to 31 days before the index date
- **Presentation**: one covariate defined for each SNOMED condition observed in the period from 30 days before the index date to the index date
- **Prior medications:** one covariate defined for each RxNorm ingredient with exposure start at least 30 days prior to the index date and exposure end on or before the index date
- **Visit context**: one covariate for the presence of an inpatient visit in the 30 days prior to the index date, and one for the presence of an emergency department visit in the same period.

Information on the prevalence of each covariate in each cohort was stored in a separate table in the database from which it was extracted and used for the computation of similarity metrics. All covariates with prevalence < 1% were excluded to avoid distortion of similarity scores by extremely low prevalence estimates.

2.4 Computation of Similarity Scores

Within each database, we formed all possible pairs of cohorts (separately for ingredient- and class-level cohorts) and computed similarity scores for each pair. We calculated the cosine similarity [14], the dot product of two vectors containing the target and comparator cohorts' covariate prevalences divided by the product of their lengths, between the two cohorts, separately for each covariate domain, and then averaged across the five domains to obtain the cohort similarity score. We then used these scores to generate ranked lists of comparators for each cohort. A cohort similarity score of 1 will occur if all covariates are the same between two cohorts, whereas a score of 0 indicates that the cohorts have no agreement across the covariates. Although other measures of multivariable similarity/distance, such as the average absolute standardized difference and the Mahalanobis distance, were available, the computational efficiency of the cosine similarity made it an attractive choice for our large-scale evaluation.

2.5 Evaluation of the Cohort Similarity Score

We evaluated the performance of the cohort similarity score in four ways. First, under the assumption that cohorts with known clinical or mechanistic relationships should have higher similarity than those without such relationships, we assessed whether drugs with closer relationships in the ATC hierarchy [15] tended to have higher cohort similarity scores than those without such relationships and whether this pattern was consistent across databases. To do so, we classified pairs of ingredient-level cohorts according to the highest ATC level at which they co-occurred, if any. Pairs that co-occurred in a class at ATC level 5 were excluded from this analysis as this level is commonly used for individual drug products or combinations.

Second, we reviewed the top five recommended comparators for six example drugs and evaluated their clinical relevance to the target qualitatively. To obtain a diverse set of examples, we selected one newer and one more established drug in each of three therapeutic areas: anticoagulation (warfarin and apixaban), rheumatoid arthritis (methotrexate and adalimumab), and multiple sclerosis (glatiramer and ocrelizumab).

Third, to describe the relationship between the cohort similarity score and a more familiar pharmacoepidemiological metric of covariate similarity, the standardized mean difference (SMD) [16], we assessed the degree of covariate balance that would result if the top recommended comparator was chosen versus the comparator ranked 500th in a random subset of 100 target cohorts in each data source. In each case, we calculated SMDs for all covariates present within the cohort pair and reported the proportion of covariates in various degrees of imbalance based on recommended cutoffs for the absolute SMD of 0.1, 0.2, 0.5, and 0.8 [17, 18].

Fourth, we searched the literature to identify high-quality observational studies employing NU designs and compared their comparator choices to our empirical estimates of similarity. We limited our search to papers published in the following journals between January 2021 and May 2023: *Science*, *Nature*, *The New England Journal of Medicine*, *JAMA*, *JAMA Internal Medicine*, *JAMA Surgery*, *JAMA Psychiatry*, *JAMA Cardiology*, *JAMA Pediatrics*, *JAMA Neurology*, *The Lancet*, *BMJ*, *Diabetes Care*, *Hypertension*, *Journal of the American College of Cardiology*, *Pharmacoepidemiology and Drug Safety*, *Drug Safety*, *Annals of Internal Medicine*, *The Lancet Digital Health*, *The Lancet Infectious diseases*, *The Lancet Diabetes & Endocrinology*, *The Lancet Neurology*, *The Lancet Oncology*, *The Lancet Respiratory Medicine*, *The Lancet Psychiatry*, and *The Lancet Global Health*. We reviewed each paper, determined whether an NU design was implemented, adjudicated the target and comparator choices to RxNorm ingredients and ATC level 4 classes, and reported the corresponding cohort similarity scores for those choices. Full details on the search parameters used and the results of the review can be found in the electronic supplementary material (ESM).

3 Results

Between 1729 and 2409 RxNorm ingredient-level NU cohorts were generated per data source. Of these, between 988 and 1359 cohorts had at least 1000 patients (Table 1). The mean number of covariates with prevalence ≥ 1% extracted per cohort was 934.5–1479.9 overall, 65.4–120.0 in the "presentation" domain, 336.9–1016.4 in the "medical history" domain, and 293.0–486.2 in the "prior medications" domain. The number of demographic and visit context covariates was roughly constant by design.

Between 487,578 and 922,761 pairs of ingredient-level cohorts with at least 1000 patients were formed per data source. Among these, similarity was lowest in the "presentation" domain (mean cosine similarity 0.213–0.490, depending on data source) and highest in the "visit context" domain (mean cosine similarity 0.869–1.000, depending on data source). Between 90 and 95% of target cohorts with at least 1000 patients had a comparator with an average cohort similarity score of ≥ 0.950. Among the six example drugs, the distribution of cohort similarity scores was consistently left-skewed, with most comparators lying between 0.7 and 0.9 (Fig. 2). Similarity was lowest among pairs of cohorts that did not occur together in any ATC classes and tended

Table 1 Summary of ingredient-level cohorts generated, and covariates extracted, by data source and covariate domain

Summary	CCAE	MDCD	MDCR	JMDC	Optum
Total number of cohorts generated					
Overall	2243	2158	2065	1729	2273
With at least 1000 patients	1359	1112	988	1082	1332
Mean (SD) number of covariates with prevalence ≥ 1% extracted per cohort with ≥ 1000 patients					
All	1030.6 (324.65)	1479.9 (432.59)	1287.7 (279.28)	934.5 (186.76)	1422.0 (441.61)
Demographics	7.8 (1.12)	9.6 (1.52)	6.2 (0.57)	8.9 (1.23)	9.3 (1.64)
Medical history	662.3 (234.73)	1016.4 (319.43)	852.1 (207.44)	336.9 (74.25)	957.3 (309.51)
Presentation	65.4 (49.17)	120.0 (63.71)	88.3 (47.32)	101.5 (36.33)	119.2 (85.42)
Prior medications	293.0 (58.56)	331.9 (67.89)	339.1 (39.66)	486.2 (83.36)	334.3 (64.94)
Visit context	1.8 (0.36)	1.9 (0.25)	1.9 (0.08)	1.0 (0.00)	1.8 (0.34)
Number of pairs of cohorts with ≥ 1000 patients	922,761	617,716	487,578	584,821	886,446
Mean (SD) similarity score among pairs of cohorts with ≥ 1000 patients					
Cohort similarity score	0.721 (0.099)	0.774 (0.106)	0.847 (0.060)	0.793 (0.090)	0.743 (0.120)
Demographics cosine similarity	0.852 (0.139)	0.846 (0.128)	0.945 (0.075)	0.857 (0.134)	0.841 (0.147)
Medical history cosine similarity	0.781 (0.137)	0.798 (0.137)	0.913 (0.059)	0.879 (0.092)	0.798 (0.141)
Presentation cosine similarity	0.213 (0.173)	0.401 (0.216)	0.490 (0.190)	0.377 (0.189)	0.349 (0.212)
Prior medications cosine similarity	0.882 (0.085)	0.872 (0.090)	0.931 (0.059)	0.917 (0.071)	0.859 (0.108)
Visit context cosine similarity	0.875 (0.150)	0.955 (0.066)	0.957 (0.069)	1.000 (0.000)	0.869 (0.160)
Number (%) of target cohorts with at least one comparator with cohort similarity score of:					
At least 0.900	1294 (95.2)	1094 (98.4)	981 (99.3)	1067 (98.6)	1294 (97.1)
At least 0.950	1087 (80.0)	961 (86.4)	924 (93.5)	1024 (94.6)	1118 (83.9)

The cohort similarity score is the average of domain-specific cosine similarities

CCAE Merative™ MarketScan® Commercial Claims and Encounters, *MDCD* Merative™ MarketScan® Multi-State Medicaid, *MDCR* Merative™ MarketScan®Medicare Supplemental, *JMDC* Japan Medical Data Center, *Optum* Optum© De-Identified Clinformatics© Data Mart, *SD* standard deviation

to increase among pairs of cohorts that co-occurred within higher-level ATC classes (Fig. 3).

The top five comparators for each of the six example drugs, by average rank across data sources (Table 2) and within data sources (Table 3) consistently corresponded to alternative treatments for each drug's indication(s). For example, the top five comparators for glatiramer (interferon beta-1a, interferon beta-1b, dimethyl fumarate, fingolimod, and natalizumab) are all other disease-modifying therapies for multiple sclerosis (MS) [19], and those ranked sixth through tenth (teriflunomide, dalfampridine, ocrelizumab, corticotropin, and modafinil; data not shown) are either disease-modifying therapies or are used to treat symptoms of MS or MS relapse. Ranked comparator lists for these example drugs varied according to data source (Figs. 2, 4A), with the highest degree of correlation between CCAE, MarketScan® Multi-State Medicaid, and Optum. An example of between-data source heterogeneity is shown in Fig. 4B where, among 627 candidate comparators for warfarin observed in both CCAE and JMDC, rivaroxaban is identified as highly similar to warfarin in both data sources, whereas xylitol and vitamin A appear much more similar to warfarin in JMDC than in CCAE, and the reverse is true for hydralazine.

Among the 100 target drugs randomly selected within each data source and their top comparators, most covariates were well balanced. When the comparator ranked 500th was chosen instead, the proportion of imbalanced covariates was higher (Fig. 5). The average proportion of covariates with $|SMD| \leq 0.1$ was between 72% and 84% for the top comparator versus between 28 and 45% for the comparator ranked 500th.

Our literature search returned 60 studies, of which 24 employed an NU design. We identified 28 distinct target–comparator pairs, of which 20 could be mapped directly to RxNorm ingredients or ATC level 4 classes. Nine such pairs were comparisons among anticoagulants, and five were among antidepressants. Among the 20 pairs that could be directly mapped, the comparator selected by the authors was in the top 5% of computed similarity for its target in at least one of our data sources in 96% of cases, and ranked among the top 10 in 91% of cases. Some pairs with notably low computed similarity were apixaban versus warfarin (cohort similarity score 0.879, rank 136 in CCAE), beta-blocking

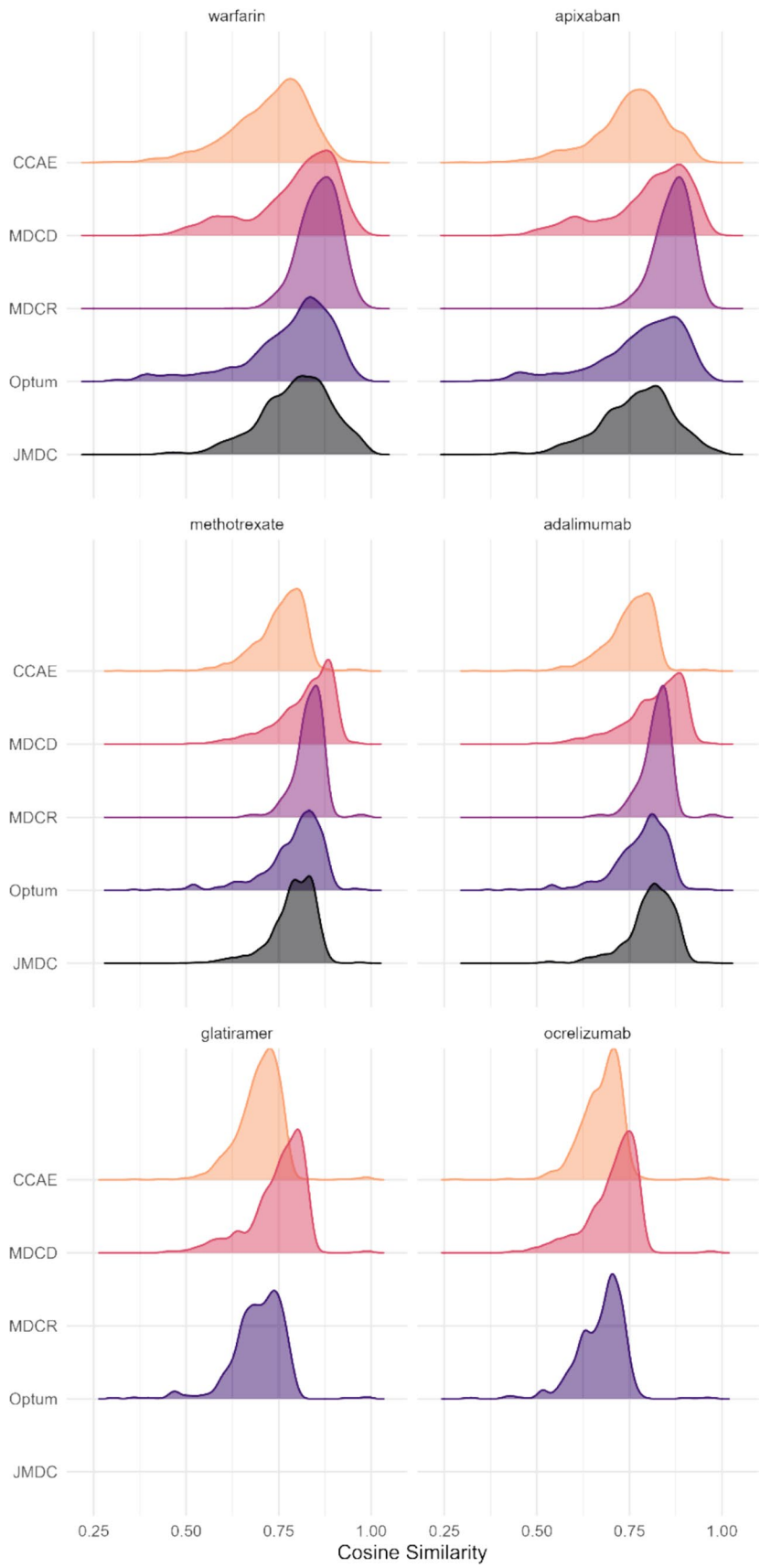

Fig. 2 Distribution of average cosine similarity scores computed across all available candidate comparators for six example drugs, by data source. Note that glatiramer and ocrelizumab did not meet the 1000 patient sample size cutoff in Merative™ MarketScan®Medicare Supplemental (MDCR) or Japan Medical Data Center (JMDC), so results are not present for these data sources. The number of available comparators ranged from 988 to 1359, depending on data source. *CCAE* Commercial Claims and Encounters, *MDCD* Multi-State Medicaid

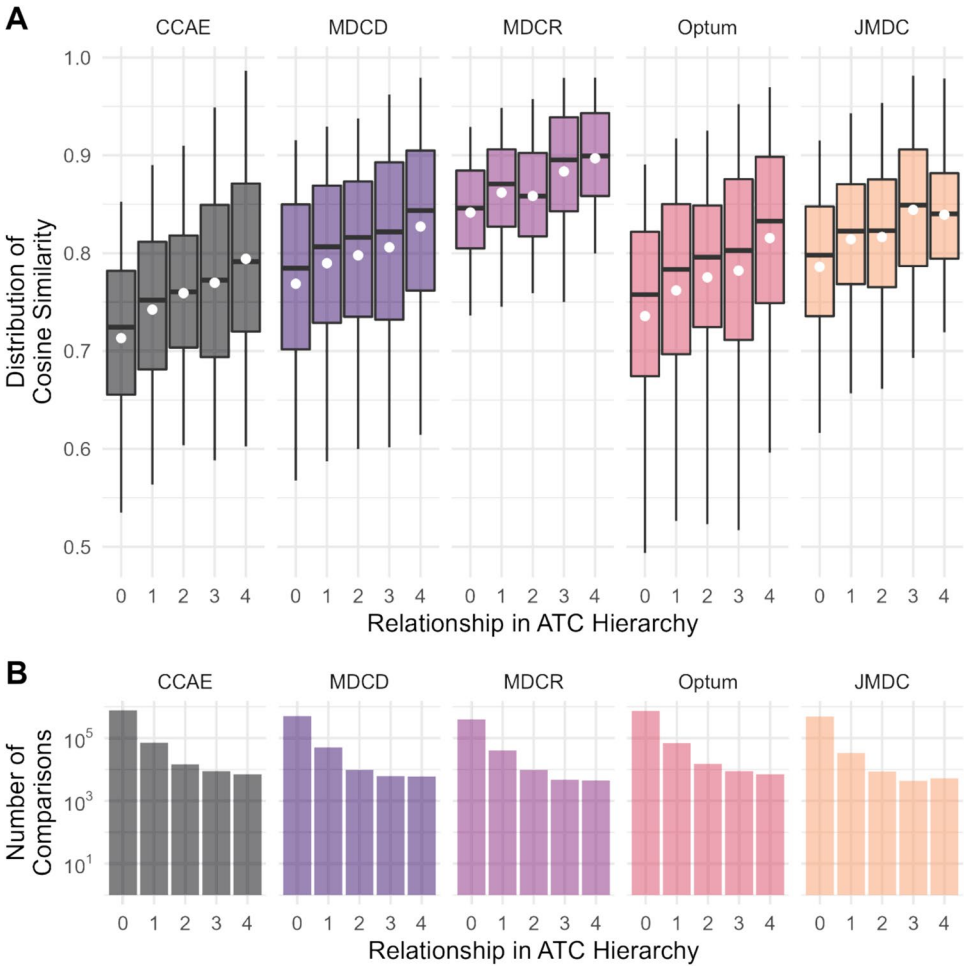

Fig. 3 Distribution of cohort similarity scores (panel **A**) and number of comparisons (panel **B**), by relatedness in Anatomic Therapeutic Chemical (ATC) hierarchy and data source. All target–comparator pairs were stratified by the distance between the two drugs in the ATC hierarchy ("4", indicating that both drugs are in same low-level class, through "0", indicating that the drugs do not share any common drug class ancestry). Cohort pairs in which the target and comparator are both members of an ATC level 5 class are omitted as these classes are frequently used for individual drugs or combination products. Whiskers are truncated at 10th and 90th percentiles. White circles indicate mean values. *CCAE* Commercial Claims and Encounters, *MDCD* Multi-State Medicaid, *MDCR* Merative™ MarketScan®Medicare Supplemental

Table 2 Top five comparators for six example drugs, according to average rank across five data sources

Rank	Warfarin	Apixaban	Methotrexate	Adalimumab	Glatiramer	Ocrelizumab
1	Digoxin (4.6; 0.960; 5)	Rivaroxaban (1.0; 0.977; 5)	Sulfasalazine (1.2; 0.978; 5)	Certolizumab pegol (2.2; 0.974; 4)	Interferon beta-1a (1.3; 0.993; 3)	Teriflunomide (1.0; 0.980; 2)
2	Rivaroxaban (6.2; 0.968; 5)	Diltiazem (7.0; 0.955; 5)	Leflunomide (2.5; 0.970; 4)	Sulfasalazine (3.4; 0.960; 5)	Interferon beta-1b (1.5; 0.993; 2)	Dimethyl fumarate (2.3; 0.969; 3)
3	Diltiazem (6.4; 0.959; 5)	Amiodarone (9.6; 0.953; 5)	Etanercept (3.4; 0.966; 5)	Infliximab (4.0; 0.958; 5)	Dimethyl fumarate (3.3; 0.991; 3)	Natalizumab (3.0; 0.967; 3)
4	Carvedilol (9.0; 0.952; 5)	Flecainide (14.2; 0.952; 5)	Adalimumab (4.6; 0.953; 5)	Methotrexate (4.2; 0.953; 5)	Fingolimod (3.3; 0.989; 3)	Fingolimod (3.7; 0.965; 3)
5	Amiodarone (10.2; 0.953; 5)	Torsemide (14.4; 0.947; 5)	Certolizumab pegol (6.5; 0.953; 4)	Leflunomide (6.0; 0.945; 4)	Natalizumab (4.3; 0.988; 3)	Dalfampridine (4.0; 0.967; 2)

Restricted to comparators occurring in at least two data sources. Entries are given as comparator name (average rank across data sources; mean cohort similarity score; number of data sources with information on comparison)

agents versus aldosterone antagonists (cohort similarity score 0.757, rank 359 in CCAE), and topiramate versus lamotrigine (cohort similarity score 0.846, rank 167 in CCAE). See the ESM for similarity scores from the reviewed literature.

4 Discussion

Over the past two decades, the NU cohort design has emerged as a best practice for the estimation of drug effects from observational data. However, the implementation of

Table 3 Top five comparators for six example drugs according to cohort similarity score, by data source

Example drug	Data Source	Comparator #1	Comparator #2	Comparator #3	Comparator #4	Comparator #5
Warfarin	CCAE	Fondaparinux (0.970)	Rivaroxaban (0.959)	Dalteparin (0.957)	Enoxaparin (0.932)	Digoxin (0.915)
Warfarin	MDCD	Diltiazem (0.973)	Carvedilol (0.970)	Dabigatran etexilate (0.970)	Rivaroxaban (0.968)	Captopril (0.968)
Warfarin	MDCR	Diltiazem (0.975)	Digoxin (0.972)	Rivaroxaban (0.971)	Amiodarone (0.971)	Sotalol (0.970)
Warfarin	Optum	Rivaroxaban (0.970)	Dabigatran etexilate (0.967)	Captopril (0.967)	Diltiazem (0.966)	Digoxin (0.965)
Warfarin	JMDC	Protamines (0.988)	Dexmedetomidine (0.986)	Norepinephrine (0.985)	Dobutamine (0.983)	Digoxin (0.982)
Apixaban	CCAE	Rivaroxaban (0.956)	Protamine sulfate (USP) (0.951)	Digoxin antibodies Fab fragments (0.940)	Diltiazem (0.940)	Hydralazine (0.935)
Apixaban	MDCD	Rivaroxaban (0.985)	Torsemide (0.967)	Diltiazem (0.967)	Bumetanide (0.966)	Carvedilol (0.964)
Apixaban	MDCR	Rivaroxaban (0.974)	Diltiazem (0.952)	Hydralazine (0.952)	Enoxaparin (0.952)	Protamine sulfate (USP) (0.951)
Apixaban	Optum	Rivaroxaban (0.973)	Digoxin antibodies Fab fragments (0.972)	Amiodarone (0.971)	Diltiazem (0.968)	Torsemide (0.966)
Apixaban	JMDC	Rivaroxaban (0.998)	Dabigatran etexilate (0.990)	Digoxin (0.988)	Edoxaban (0.985)	Flecainide (0.985)
Methotrexate	CCAE	Sulfasalazine (0.972)	Leflunomide (0.970)	Etanercept (0.969)	Hydroxychloroquine (0.956)	Adalimumab (0.955)
Methotrexate	MDCD	Sulfasalazine (0.961)	Etanercept (0.960)	Hydroxychloroquine (0.955)	Adalimumab (0.954)	Leflunomide (0.950)
Methotrexate	MDCR	Leflunomide (0.984)	Sulfasalazine (0.984)	Hydroxychloroquine (0.981)	Adalimumab (0.980)	Etanercept (0.979)
Methotrexate	Optum	Sulfasalazine (0.983)	Leflunomide (0.976)	Hydroxychloroquine (0.974)	Etanercept (0.958)	Adalimumab (0.955)
Methotrexate	JMDC	Sulfasalazine (0.988)	Tocilizumab (0.968)	Etanercept (0.966)	Golimumab (0.962)	Adalimumab (0.922)
Adalimumab	CCAE	Certolizumab pegol (0.973)	Etanercept (0.962)	Sulfasalazine (0.957)	Methotrexate (0.955)	Infliximab (0.948)
Adalimumab	MDCD	Sulfasalazine (0.970)	Certolizumab pegol (0.970)	Etanercept (0.960)	Methotrexate (0.954)	Infliximab (0.942)
Adalimumab	MDCR	Etanercept (0.990)	Leflunomide (0.983)	Infliximab (0.982)	Methotrexate (0.980)	Certolizumab pegol (0.978)
Adalimumab	Optum	Certolizumab pegol (0.973)	Etanercept (0.960)	Sulfasalazine (0.959)	Methotrexate (0.955)	Golimumab (0.948)
Adalimumab	JMDC	Infliximab (0.982)	Azathioprine (0.963)	Golimumab (0.956)	Sulfasalazine (0.939)	Methotrexate (0.922)
Glatiramer	CCAE	Interferon beta-1a (0.994)	Interferon beta-1b (0.993)	Fingolimod (0.992)	Natalizumab (0.992)	Dimethyl fumarate (0.991)
Glatiramer	MDCD	Interferon beta-1a (0.994)	Dimethyl fumarate (0.992)	Fingolimod (0.987)	Natalizumab (0.986)	Ocrelizumab (0.962)
Glatiramer	Optum	Interferon beta-1b (0.992)	Interferon beta-1a (0.991)	Dimethyl fumarate (0.989)	Fingolimod (0.989)	Natalizumab (0.987)
Ocrelizumab	CCAE	Teriflunomide (0.981)	Dimethyl fumarate (0.974)	Fingolimod (0.969)	Natalizumab (0.968)	Peginterferon beta-1a (0.965)
Ocrelizumab	MDCD	Natalizumab (0.977)	Dimethyl fumarate (0.975)	Fingolimod (0.974)	Glatiramer (0.962)	Interferon beta-1a (0.953)
Ocrelizumab	Optum	Teriflunomide (0.979)	Dalfampridine (0.971)	Dimethyl fumarate (0.960)	Natalizumab (0.957)	Fingolimod (0.952)

Entries are given as comparator name (cohort similarity score). Note that glatiramer and ocrelizumab did not meet the 1000 patient sample size cutoff in MDCR or JMDC, so results are not present for these data sources

CCAE Merative™ MarketScan® Commercial Claims and Encounters, *MDCD* Merative™ MarketScan® Multi-State Medicaid, *MDCR* Merative™ MarketScan®Medicare Supplemental, *JMDC* Japan Medical Data Center, *Optum* Optum© De-Identified Clinformatics© Data Mart

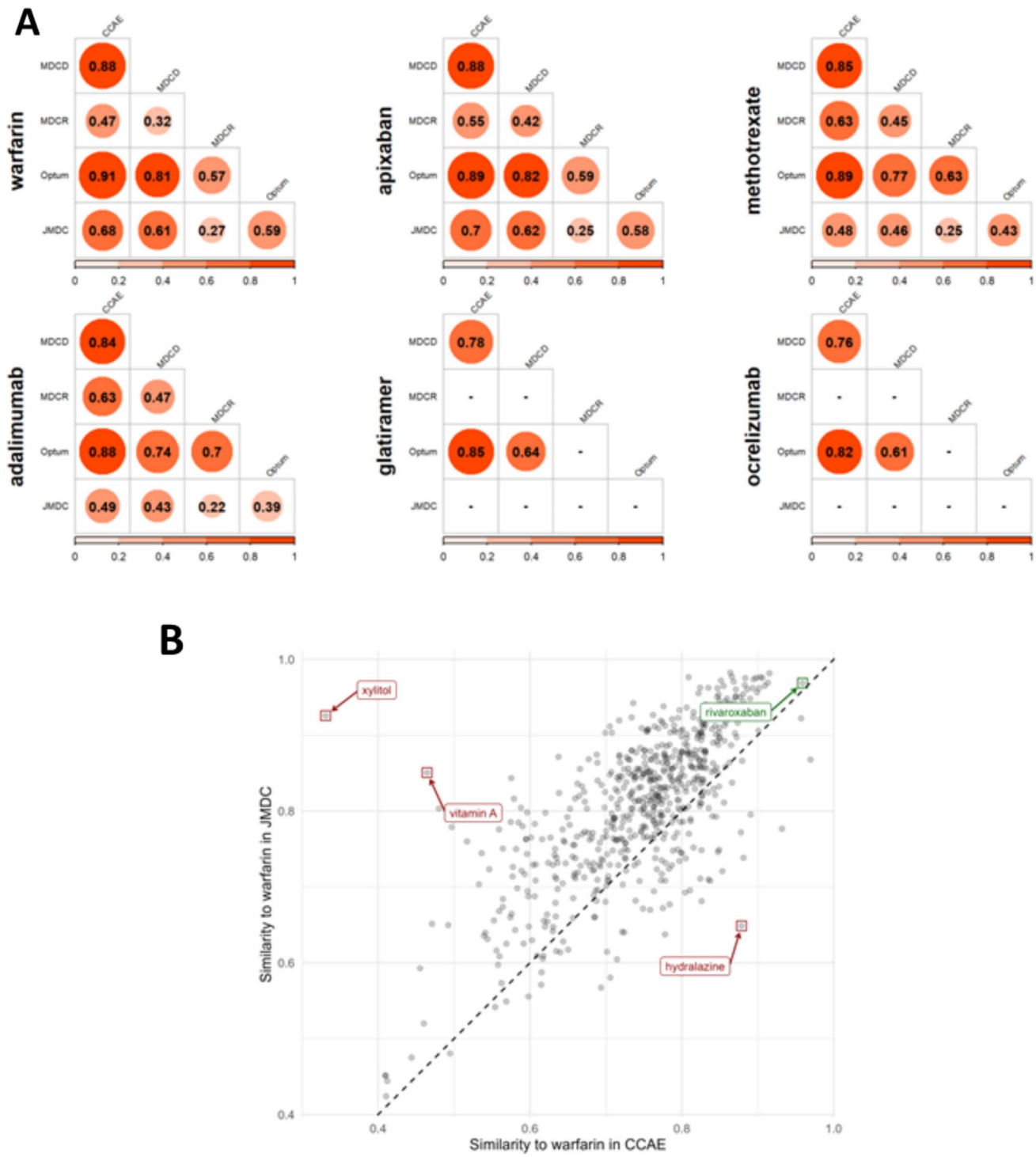

Fig. 4 Pairwise rank correlations in cohort similarity scores between data sources for six example drugs (**A**) and cosine similarity scores among 627 candidate comparators for warfarin observed in both Commercial Claims and Encounters (CCAE) and Japan Medical Data Center (JMDC) (**B**). *MDCD* Multi-State Medicaid, *MDCR* Merative™ MarketScan®Medicare Supplemental

this design may be hampered by the difficulty of selecting an appropriate comparator for the target drug of interest, either because there are many potential alternatives to the target drug or because the true target of inference is in fact a contrast between the target drug and its absence. The development of empirical comparator recommendations can aid investigator decision-making, simplifying the process of comparator selection and improving the quality of

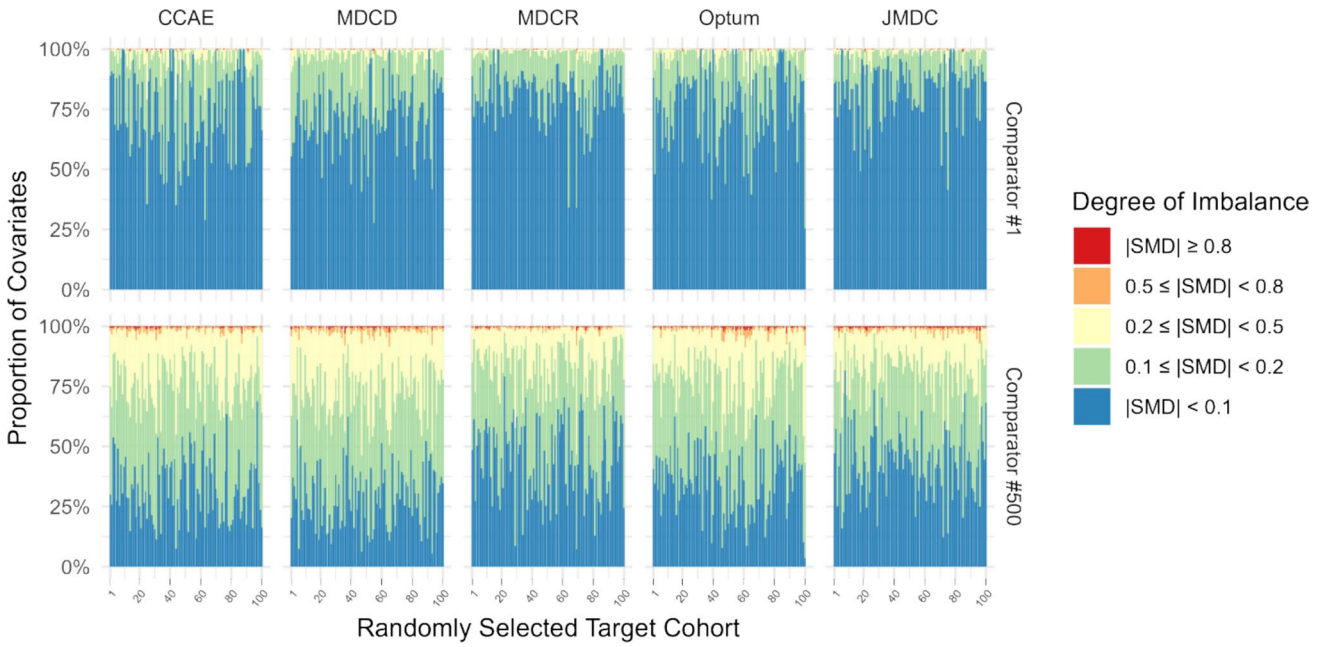

Fig. 5 Proportion of covariates at various levels of imbalance according to the standardized mean difference (SMD) among 100 randomly selected target cohorts and their comparators ranked 1st and 500th, by data source. *ATC* Anatomic Therapeutic Chemical, *CCAE* Commercial Claims and Encounters, *JMDC* Japan Medical Data Center, *MDCD* Multi-State Medicaid, *MDCR* Merative™ MarketScan®Medicare Supplemental

pharmacoepidemiological research and the speed of post-marketing surveillance. Ultimately, these approaches may even enable the automation of evidence generation using the NU design, something that was previously possible only with self-controlled designs [20, 21] and temporal scans [22, 23].

We demonstrated a method for ranking potential comparators based on similarity in high-dimensional covariate space. We showed that most target cohorts of reasonable size possessed at least one potential comparator with a high degree of similarity and that cohorts that were more related in the ATC hierarchy also had higher cohort similarity scores. For a random sample of 100 cohorts within each data source, we also showed that selecting the top-most similar comparator identified a cohort with balance on most covariates before any adjustment took place.

We additionally demonstrated that, for six target cohorts from a diverse set of therapeutic areas, the top-ranked comparators aligned well with subject matter knowledge. That is, all were used in the treatment of at least one of the target cohort's indications. Indeed, many of these top-ranked comparators are in use in head-to-head safety or effectiveness comparisons with the target cohort in the literature or in public registries of clinical and observational studies. For example, head-to-head studies exist comparing apixaban and its top comparator rivaroxaban [24, 25], adalimumab and its top comparator certolizumab [26, 27], and ocrelizumab and its top comparator teriflunomide [28]. This was also borne out in our literature review, which demonstrated that NU studies published in high-quality journals tended to select comparators that our method also ranked highly. This is reassuring, but it should also be noted that the correlation in comparator rankings across data sources was not consistently high (Fig. 4). This may indicate that a given comparator is not uniquely fit for purpose across all data sources—perhaps due to differences in covered treatments, reimbursement patterns, or variations in local treatment strategies—and, as such, investigators should take care to select a comparator using information from a source that at least resembles their intended analytic context or to implement our methodology in their own data.

This study complements a large body of work on drug similarity from the bioinformatics community, including attempts to define similarity based on chemical structure, gene expression [29], genome-wide associations [30], co-occurrence of drugs with condition terms in the literature [31], individual patient data [32], and a combination of these. However, our work is unique in its focus on the use of similarity indices for comparator selection rather than for identification of novel associations and for its clinically oriented approach to covariate data extraction. The empirical comparator recommendations provided here have several advantages. First, they can reduce subjectivity, both in terms of which drugs may serve as candidate comparators for a given target drug and which covariates should be used to determine empirical feasibility. Although comparator

choice should be guided by subject matter knowledge [33], no investigator is omniscient, so the availability of empirical recommendations may promote more informed decisions and could potentially enable less biased comparisons. Additionally, empirical recommendations may reveal previously unknown comparators that are not necessarily alternative treatments for any of the target drug's indications. They thus hold promise for the identification of appropriate inactive comparators/negative control exposures, ultimately allowing for cohort studies that emulate user versus non-user comparisons within the NU framework. Additionally, these recommendations have been assessed across multiple large data sources comprising diverse populations within and outside the USA. They have been observed, consistently across these data sources, to label drugs that are more closely related in the ATC hierarchy as more similar, to produce balance on a majority of covariates when the most similar comparator is chosen, and to identify top comparators that align with subject matter knowledge and high-quality published research. Finally, the approach presented here provides a standardized framework for comparator ranking that can be readily implemented by investigators. This standardization can improve reproducibility and scalability and ultimately lowers the burden on domain expertise for the investigator.

Compared with traditional approaches, a limitation of the approach introduced here is that the cohort similarity score is based on the *marginal* distributions of covariates in two cohorts and as such is only a proxy for full covariate similarity. In contrast, propensity score methods have the advantage of providing metrics (e.g., the c-statistic or the preference score [34]) that reflect the similarity in *joint* covariate distributions across cohorts. Furthermore, such methods can be used to adjust for observed dissimilarity when it occurs. However, unlike many propensity score models, which only incorporate a small number of covariates, the cohort similarity score dynamically defines a large set of covariates based on all available data elements. It should be noted that this data-adaptive approach has been incorporated into some propensity score methods such as the high-dimensional propensity score [35] and, more recently, the large-scale propensity score (LSPS) [36]. However, the cohort similarity score remains the only approach that is scalable to millions of candidate target–comparator combinations, whereas propensity score models—particularly those using high-dimensional covariate data—can be computationally expensive, making them infeasible for screening large numbers of candidate comparators. For example, on an Amazon EC2 Instance running Windows Server 2019 with a 2.2 GHz processor and 64 GB RAM and accessing data stored in a Redshift database, our entire pipeline (including generating cohorts, extracting covariate data, and computing cohort similarity scores for all drugs) took between 33 and 172 minutes per database, whereas an LSPS pipeline (including generating cohorts, extracting covariate data, and fitting models) took between 3 and 40 hours *for a single target–comparator pair*. The cohort similarity score may therefore be best suited as a tool to narrow the comparator search space before techniques such as the high-dimensional propensity score or LSPS are employed. It should also be noted that, like propensity score methods and assessments of standardized differences in covariates, our approach only addresses part of the confounding equation. That is, it summarizes covariate imbalance without addressing potential relationships between covariates and study outcomes. Although investigators must keep this limitation in mind, they can address it post hoc by searching our web application to identify the degree of balance on important risk factors for their study outcome of interest.

Several factors may limit the usefulness of the comparator recommendations provided here. First, comparator recommendations for drugs that are used in different populations depending on dose, form, or route of administration may be less coherent than recommendations for drugs that are used more consistently regardless of these factors. The same limitation may apply to drug ingredients that are used in combination preparations or that possess multiple, possibly unrelated, indications. However, this can be ameliorated by running the entire analytic pipeline in the subset of the population with a specific indication of interest. A demonstration of this for the target drug spironolactone, which is used in the treatment of both acne and heart failure, showed that such restriction both reduces the comparator search space (Fig. 1 in the ESM) and improves the coherence of recommendations (Table 1 in the ESM). Although such restriction is not feasible at the scale of all indications, it may be attractive to investigators whose work routinely focuses on a small set of indications. Second, our recommendations may be less useful in situations where the treatment landscape for a given indication has changed dramatically over time or when an investigator wishes to restrict their study to a particular calendar period. For example, the top ranked comparators for the rheumatoid arthritis treatment adalimumab include several older drugs such as methotrexate, leflunomide, and sulfasalazine, whereas an analysis restricting cohort generation to a more recent period may preference newer biologic treatments that are now used more frequently as first-line therapies. Future research could explore the impact of including calendar time restriction in the cohort definition/generation process and/or the inclusion of calendar time-based covariates in similarity calculations on the rankings. Third, the assessment of these recommendations was limited by the lack of an objective metric for defining

comparator "appropriateness." Although we sought to benchmark appropriateness based on comparator location in the ATC hierarchy and appearance in published literature or registered studies, future work might attempt to go further by demonstrating that the use of empirical comparator recommendations can lead to less biased treatment effect estimates than traditional methods of comparator selection.

Finally, it should be clearly stated that empirical comparator recommendations are not a replacement for critical thinking in study design and analysis. Investigators may wish to use them as a "first pass" to identify and evaluate candidate comparators, after which more careful consideration and detailed analysis can determine true feasibility. Crucially, investigators must remain aware that covariate adjustment is almost invariably required, regardless of the quality of the comparator chosen. Similarity between a target drug and comparator is also no guarantee that any remaining covariate imbalance can be successfully adjusted via established methods such as restriction or propensity score adjustment. Conversely, low similarity relative to other recommendations is not a guarantee that successful adjustment is impossible.

Our work leaves open many exciting avenues for further exploration of empirical comparator recommendations. In particular, work may interrogate whether the covariate data extraction process could be altered to improve similarity calculation. For example, by adding richer demographic or socioeconomic covariate data where sources allow, by varying the time windows during which covariates are assessed, or by restricting on/adjusting for calendar time. Alternate approaches to validation, such as comparing the proposed method with more computationally expensive techniques such as propensity score overlap assessment, could also be explored. Future work might also explore the utility of adding sources of metadata on known or suspected drug–outcome relationships, such as the Observational Health Data Sciences and Informatics (OHDSI) Common Evidence Model [37, 38], to improve the ease of identifying inactive comparators/negative control exposures.

5 Conclusion

We have developed and evaluated a method for recommending comparators that yields promising results and that may be of use to investigators wishing to conduct NU studies of drug safety and effectiveness. An open-source tool set is available for those wishing to identify candidate comparators using their own data, and the full suite of results, including comparator rankings for thousands of drugs across multiple data sources, is available to the public. We hope these tools and the results presented here will accelerate and improve the quality of future pharmacoepidemiologic research.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s40264-025-01569-y.

Declarations

Funding This work was funded by Janssen Research and Development, a subsidiary of Johnson & Johnson. All authors are employees of Janssen Research and Development.

Conflicts of interest All authors are employees of Johnson & Johnson.

Consent for publication Not applicable.

Consent to participate Not applicable.

Ethics approval Not applicable.

Data availability The full suite of results described here is available as an interactive web application (https://data.ohdsi.org/ComparatorSelectionExplorer), which can be used to identify and diagnose potential comparators for a given target cohort. Note that this application undergoes periodic updates, including the updating of existing data sources to newer versions, the addition of new data sources, and the modification of existing functionality. As such, some results derived from the application may differ from those presented here.

Code availability All code necessary to run the analytic pipeline described here and to generate the web application is available via GitHub (https://github.com/OHDSI/ComparatorSelectionExplorer).

Author contributions Justin Bohn designed and performed the analysis, conducted the literature review, drafted the manuscript, and led the manuscript revision process. James P. Gilbert designed and performed the analysis and reviewed and contributed to the manuscript. Christopher Knoll designed the analysis and reviewed and contributed to the manuscript. David M. Kern designed the analysis and reviewed and contributed to the manuscript. Patrick B. Ryan designed the analysis, conducted the literature review, and reviewed and contributed to the manuscript. All authors read and approved the final version.

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